An unbounded cost of continuous decoding on Banach unit balls

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Abstract

For the fixed inclusion $\ell_1([0,1]) \hookrightarrow \ell_2([0,1])$, continuous measurements with arbitrary decoding admit errors tending to zero, while continuous decoding has optimal error one at every finite measurement dimension. Thus no universal continuity-cost constant exists for arbitrary real Banach spaces.

Scope and status. The counterexample uses nonseparable Banach spaces, as permitted by AIM 570. The corresponding question restricted to separable spaces is not settled here. This is a solution claim submitted for review. Pre-submission mathematical, bibliographic, and editorial checks were performed with OpenAI Codex (AI); independent human review and formal verification are not claimed.

1 Problem and proof

The quantification in entry 570 is over arbitrary real Banach spaces, with no separability assumption [3]. For a bounded linear $S : X \to Y$, write

$$e_n(S) = \inf_{N,\phi} \sup_{x \in B_X} \|Sx - \phi(N(x))\|_Y ,$$

where $N : B_X \to \mathbb{R}^n$ is continuous for the norm topology and $\phi : \mathbb{R}^n \to Y$ is arbitrary. Define $\delta_n(S)$ by the same infimum with ϕ continuous on all of $\mathbb{R}^n$. Both quantities optimize over the encoder as well as the decoder; the encoding is nonadaptive. The question appears after Proposition 9.5 of Krieg–Ullrich [2]. The compact-input factor-two theorem in Lemma 9 of Krieg–Novak–Ullrich [1] does not apply to the following unit ball.

Theorem 1.1. *Let*

$$I = [0,1], \qquad X = \ell_1(I), \qquad Y = \ell_2(I),$$

and let $S : X \to Y$ be the canonical inclusion. Then, for every finite $n \geq 1$,

$$\delta_n(S) = 1,$$

whereas for every integer $K \geq 1$,

$$e_{2K}(S) \leq \frac{1}{\sqrt{K+1}}.$$

Therefore no universal finite constant C can satisfy $\delta_n(S) \leq C e_n(S)$ in the stated class. In particular, $C = 2$ already fails for this operator at $n = 8$.

1.1 A continuous encoder with a discontinuous accurate decoder

Every element of $\ell_1(I)$ or $\ell_2(I)$ has countable support. The inclusion S is bounded with norm one. Fix $K \geq 1$ and put $\tau = 1/(K+1)$. Define coordinatewise soft thresholding by

$$(T_\tau x)_t = \operatorname{sgn}(x_t)(|x_t| - \tau)_+.$$

This map is 1-Lipschitz on $\ell_1(I)$. If $\|x\|_1 \leq 1$, then $T_\tau x$ has at most K nonzero coordinates: $K+1$ coordinates strictly larger than τ in magnitude would have total ℓ_1 norm greater than one. Also,

$$\|x - T_\tau x\|_2^2 = \sum_{t \in I} \min(|x_t|, \tau)^2 \leq \tau \sum_{t \in I} |x_t| \leq \tau. \tag{1}$$

Use $2K$ real moments of the thresholded vector as the encoder:

$$N_K(x)_j = \sum_{t \in [0,1]} t^j (T_\tau x)_t, \qquad j = 0, 1, \ldots, 2K-1. \tag{2}$$

Here $t^0 = 1$, including at $t = 0$. Every coordinate is continuous, indeed 1-Lipschitz in the ℓ_1 norm, because $|t^j| \leq 1$. Thus $N_K : B_X \to \mathbb{R}^{2K}$ is an admissible nonadaptive continuous encoder.

The moment map in (2) is injective on vectors supported on at most K indices. To prove this, subtract two such vectors with the same moments. Their difference has support $\{t_1, \ldots, t_r\}$, where $r \leq 2K$, and its coefficients c_i satisfy

$$\sum_{i=1}^{r} c_i t_i^j = 0, \qquad j = 0, \ldots, r-1.$$

The Vandermonde matrix of distinct nodes t_i is invertible; hence all $c_i = 0$.

Define a decoder on all of $\mathbb{R}^{2K}$ by returning the unique at-most-K-sparse vector having the given moments whenever one exists, and returning zero otherwise. No continuity is required for e_{2K}. On the encoder image it returns $T_\tau x$ exactly. Equation (1) proves

$$e_{2K}(S) \leq \sqrt{\tau} = (K+1)^{-1/2}.$$

1.2 Every continuous decoder misses a coordinate

Let $\phi : \mathbb{R}^n \to \ell_2(I)$ be any continuous decoder. Its image is separable, since $\mathbb{R}^n$ is separable. More explicitly, choose a countable dense subset of its image and let $J \subset I$ be the union of the supports of those countably many vectors. Each support is countable, so J is countable. Since $\ell_2(J)$ is closed in $\ell_2(I)$,

$$\phi(\mathbb{R}^n) \subset \ell_2(J).$$

Choose $t \in I \setminus J$ and take the unit vector $e_t \in B_X$. For every encoder N, regardless of its form,

$$\|Se_t - \phi(N(e_t))\|_2^2 = 1 + \|\phi(N(e_t))\|_2^2 \geq 1.$$

Thus $\delta_n(S) \geq 1$. The zero decoder gives the reverse inequality, because $\|S\| = 1$.

For any putative universal C, choose $K + 1 > C^2$. Then

$$Ce_{2K}(S) \leq \frac{C}{\sqrt{K+1}} < 1 = \delta_{2K}(S).$$

For $C = 2$, take $K = 4$, giving $2e_8(S) \leq 2/\sqrt{5} < 1$.

Remark 1.2. The same fixed operator supplies the counterexample for every proposed constant. Both its domain and codomain are nonseparable. No conclusion about the still-interesting variant restricted to separable Banach spaces is claimed.

References

[1] D. Krieg, E. Novak, and M. Ullrich, *How many continuous measurements are needed to learn a vector?*, arXiv:2412.06468v2, 22 September 2026. Lemma 9 and the discussion after Theorem 10.

[2] D. Krieg and M. Ullrich, *Approximation of functions: Optimal sampling and complexity*, Acta Numerica **35** (2026), 273–457. doi:10.1017/S0962492925100287. Proposition 9.5 and the question on page 411.

[3] M. Colbrook, AIM catalogue, entry 570: The cost of continuous reconstruction on Banach unit balls. Repository statement accessed 3 October 2026.